

ARC-AGI1 ALGORITHM

(POSSIBLY WRONG) INFORMAL NOTES BEFORE ARRIVING AT A PRECISE FORMULATION

1. INTERACTING ENTITIES AND INTERACTIONS OF HiS-ARC

- (1) Start writing (Issue #2) the list of geometric entities, geometric relations, geometric properties and geometric transformations shown in the lecture.
- (2) Mrs Code is now called HiS-ARC (**H**uman **I**nspired **I**S for **A**bstract **R**easoning **C**orpora).
- (3) In order to implement an efficient selection of geometrical objects, construct a hierarchical human-inspired organization of HiS-ARC as an IS:

(a) Human-inspired:

- (i) **Synaptic plasticity**: based on the idea of interactions' good measuring "how many times we used this interacting entity/interaction so far" and normalized into a probability of selection of this interacting entity/interaction in the algorithm (=: dynamics of HiS-ARC as a dynamical IS). See below Sec. 3.
- (ii) **Abstract geometric entities**: in the IS corresponding to HiS-ARC, consider interactions of the form e.g.

$$\begin{aligned}
 i : \text{rectangles} &\xrightarrow{r} \text{closed lines} \\
 j : \text{triangles} &\xrightarrow{s} \text{closed lines} \\
 k : \text{squares} &\xrightarrow{u} \text{rectangles} \\
 h : \text{L-shapes} &\xrightarrow{v} \text{L-shapes},
 \end{aligned}$$

where the propagators have the following properties:

- (A) $ac_i^r(t) = 1$ means that at time t (computation time) we have found a 'rectangle' in at least one training $\mathcal{I}_k$ but it occurs also in the others $\mathcal{I}_j$ only if we think at it as a more abstract "closed line" (e.g. because in $\mathcal{I}_j$ there is a 'triangle'). See also below 1. Similarly for $\mathcal{S}_k$.
 - (B) $\gamma_i(t)$ is a counting good following *synaptic plasticity*, i.e. "how many times we used this abstraction so far", see Sec. 3.
 - (C) An interaction of the type $h : g \xrightarrow{v} g$, where g is an abstract geometrical objects, simply means that g occurs in some $\mathcal{I}_k$.
 - (D) Also geometric relations and transformations have to be organized in a hierarchical way using a similar abstraction, like geometric entities as above.
- (iii) **Abstract geometric relations between geometrical entities**: These interactions generalize the list of geometric relations. For example:

$$\begin{aligned}
 j : \text{cells} &\xrightarrow{r} \text{rectangle} \\
 k : \text{rectangles, square} &\xrightarrow{s} \text{lines},
 \end{aligned}$$

with $j.\text{pr.type} = \text{'contains'}$, $k.\text{pr.type} = \text{'share a line'}$, and γ_j, γ_k are "counting goods" as above.

- (iv) **Abstract geometric transformations**: These interactions corresponds to the list of geometric transformation. For example:

$$h : \text{lines, cells} \xrightarrow{u} \text{lines},$$

with $h.\text{pr.type} = \text{'change the color of the 1st using the color of the 2nd'}$, and γ_h as above.

- (4) Consider only interacting entities and interactions of HiS-ARC that occur in **all** $\mathcal{I}_k$ **or** in **all** $\mathcal{S}_k$. For example, a geometric entity g "occur" in all $\mathcal{I}_k$ means that an abstraction, as described in 3(a)ii, of minimal level occurs in all $\mathcal{I}_k$. For example, if a 'rectangle' occurs in $\mathcal{I}_1$ and $\mathcal{I}_2$, and a 'square' in $\mathcal{I}_3$, we have an occurrence of 'rectangle'. If we have 'rectangle' in $\mathcal{I}_1$, a 'triangle' in $\mathcal{I}_2$ and a 'circle' in $\mathcal{I}_3$, then we have an occurrence of 'closed lines'. This would correspond to a certain application of the GEP in HiS-ARC in

the dynamics of **His-ARC** as an IS. However, since it is a strict rule, it would correspond to maximum cost where $C_y^j(e) \in \{0, 1\}$ (if e.g. $e \in \mathcal{I}_k \setminus \mathcal{I}_h$) or zero probability $Q_\gamma(i) = 0$ (if e.g. the interaction $i \in \mathcal{S}_k \setminus \mathcal{S}_h$). Up to now, we can therefore not use the GEP but simply apply this strict rule.

2. CEF

- (1) Differently to what I stated during the lecture, it seems more natural to consider that from the training pair $\mathcal{I}_k - \mathcal{S}_k$ we are constructing two CEF for RLE, one of which is a forgetful one (because some interacting entities are not considered at all in the solution). Moreover, in general some geometric transformations to construct $\mathcal{S}_k$ from $\mathcal{I}_k$ needs *more than one* interacting entity, such as: copy on $\mathcal{S}_k$ the rectangle in $\mathcal{I}_k$ but with a new color taken from the most numerous set of cells in $\mathcal{I}_k$. *Since we start from more than one interacting entity* (the rectangle and its interior cells), the natural learning is RLE.
- (2) On the contrary, in transforming the final empty $\mathcal{S}_N$ for the solution, we proceed stepwise starting from one interacting entity to the next one, and aiming to change its state:
 - (a) Change the dimension of the grid by following the rules used in the training (the grid is surely an interacting entity of **His-ARC**).
 - (b) Change the color of this part of the grid by following the rules used in the training.
 - (c) etc.

In this case, the natural learning process is therefore RLS because we start *from only one* interacting entity and we want to change its state.

- (3) In considering CEF constructed as composition of at most 3 elementary transformations, we do not have to consider all possible triples of transformations, but only those that are composable (i.e. the output of the first transformation can be considered as input of the subsequent one).
- (4) It could be useful to see CEF as interacting entities of another IS within **His-ARC**. In this way, we can talk of the state of a CEF, one of whose component could be its “abstract description”. This state can be used to apply the strict rule of considering the same transformations in all the pair $\mathcal{I}_k - \mathcal{S}_k$.
- (5) Visualize a CEF $U(x) = y$ by changing the color of the border (1 pixel width) of each interacting entities: first x and after y .
- (6) Make disappear all the interacting entities that are forgotten ($U(x) = \emptyset$) by a CEF.

3. ORGANIZATION OF **His-ARC** AS IS

- (1) **His-ARC** is implemented as a dynamical (i.e. not static) IS having:
 - (a) All the found geometric objects of the training IS are included in **His-ARC** as interacting entities or interactions.
 - (b) **Synaptic plasticity algorithm following the GEP**: This algorithm aims to solve the problem “how to use a human-inspired algorithm in order to find what geometric entities, relations or transformations occur in the given training IS avoiding a brute-force algorithm”. It uses the GEP, so it is implemented as a CAS.
 - (i) At t_{st} all the counting goods $\gamma_{i_1}, \dots, \gamma_{i_N} = 1$. The interpretation is: **His-ARC** already has a human-inspired core-knowledge (we can even call it *subconscious knowledge*) which allows it to recognize the geometric object i_k (entity, relation, transformation). However, initially at $t = t_{st}$ **His-ARC** has no experience about how frequently i_k occurs in ARC-AGI1 problems, but it will adapt its behaviour in learning to solve these puzzles (this is plasticity, i.e. adaptation based on occurring frequencies of objects i_k , see “Gemini - synaptic_plasticity.pdf” in Doc).
 - (ii) At generic step: By normalization, transform all the counting goods into a probability.
 - (iii) Extract a geometrical object i_k using this probability.
 - (iv) Search for occurrences of this object in the training IS.
 - (A) In the language of IS, searching for the geometric object g corresponds to activate an interaction $i_k : g \xrightarrow{v} g$ as described in **3(a)ii** in Sec. 1 (see below **1(b)ivC**).
 - (B) If i_k occurs in all the training IS, **His-ARC** finds if some i_j of lower abstraction level (see **3(a)ii** in Sec. 1) occurs in the training IS, until it finds the lower abstraction level (e.g. it could happen that the firstly extracted i_k is ‘closed lines’ whereas at the end the lower abstraction level i_j is ‘rectangles’).
 - (C) The found lowest abstraction level corresponds to the interaction $i_j : g \xrightarrow{v} g$ as described in **3(a)ii**, Sec. 1, i.e. we can say that $ac_{i_j}^v(t) = 1$. Increase its counting goods: $\gamma_{i_j}(t+1) = \gamma_{i_j}(t) + 1$ (t is not actually implemented in the code and intuitively corresponds to “next step in the algorithm”). Drop i_j from the list $i_1, \dots, i_N$ and restart from **1(b)iii**.

- (D) If i_k does not occurs in all training IS, **His-ARC** searches for occurrences at a higher abstraction level.
- (E) The highest found abstraction level is one of the i_j . Increase its counting goods: $\gamma_{i_j}(t+1) = \gamma_{i_j}(t) + 1$. Drop i_j from the list $i_1, \dots, i_N$ and restart from **1(b)iii**.
- (F) If no higher level occurs in all training IS, then drop i_k from the list $i_1, \dots, i_N$ and restart from **1(b)iii**.
- (G) In the language of IS and the GEP, this corresponds to $Q_{\gamma_{i_k}(t+1)}(i_k) = 0$ in the diversification probability or deactivation of of the propagator $\text{pr}(i_k)$, $\text{ac}_{i_k}^{\text{pr}(i_k)}(t+1) = 0$, and setting a maximal value of cost $C_{y(t+1)}^h(g) = 1$, where $i_k : g \xrightarrow{v} g$ and the global state $y(t+1)$ contains the new state $\text{ac}_{i_k}^{\text{pr}(i_k)}(t+1) = 0$.
- (v) With the new counting goods, restart from **1(b)ii**.

It is possible to prove that, at each step in this synaptic plasticity algorithm, the normalized probability is actually the maximum entropy probability on $i_1, \dots, i_N$ that “respects the past **His-ARC** experience”, i.e. the counting goods experienced by **His-ARC** in the past. In fact this probability maximize the entropy among all probabilities $p_1, \dots, p_N$ such that $p_k(t) \geq \frac{\gamma_{i_k}(t)}{\sum_{j=1}^N \gamma_{i_j}(t)} =: P_k(t)$ for all the i_k that **His-ARC** already found in the past, $k = 1, \dots, M \leq N$. Normalization also corresponds to assign “the remaining probability” $p_j(t) = \frac{1 - \sum_{k=1}^M P_k(t)}{N - M}$ to all not-founded objects $j = M + 1, \dots, N$. Therefore, this synaptic plasticity algorithm (is very simple and natural and) satisfies the GEP (i.e. it implement synaptic plasticity as a CAS).

- (c) **Object affordance algorithm** (see “Gemini - Object Affordance & Motor Priming.pdf” in Doc):
 - (i) Recognize occurrences of interacting entities in each training IS $\mathcal{I}_k, \mathcal{S}_k$.
 - (ii) Select only geometric relations and geometric transformations having these interacting entities as agents or patients (once again, this selection is an application of the GEP, setting to zero the probability to have interactions with other entities).
 - (iii) Using synaptic plasticity, recognize what relations and transformations occur in the training IS.
- (d) Other interactions (See also issue #6 and “Gemini - geometric_hierarchy_brain_processing.pdf”): The idea is to recognize forms by using “composition of elementary geometrical objects and relations”. Example: even if we do not have a code for the recognition of rectangles, we can say that a rectangle is “an L shape (or a certain form) touching a vertical line touching an L shape (of another form) touching an horizontal line... touching the initial L shape”. Here the elementary geometrical objects and relations are “L shape”, “touching” “counting and differentiating geometrical objects”, “vertical line”, “horizontal line”. We can hence construct CEF by linking these elementary objects and their interactions. This could be a general method when we consider a puzzle that is not solvable because the core-knowledge we developed so far is incomplete. It is probably the method used by human’s brain, see “Geon Theory for 2D Shapes”, “Biederman’s alphabet of 2D edge properties” and “Structural Information Theory (SIT) by Emanuel Leeuwenberg”. We can try this idea if the final percentage of solved puzzle is too low because of incomplete core-knowledge. Starting initial ideas for the formalization of this method using the language of IS are as follows:

- (i) **Human-inspired hierarchical recognition of interacting entities:** $a_1, \dots, a_n \xrightarrow{r} p$ if the geometric entities $a_1, \dots, a_n$ are used to recognize the geometric entity p . For example:

$$i : \text{lines, L shapes} \xrightarrow{r} \text{rectangles}.$$

The propagator r contains counting goods as in the synaptic plasticity described above, i.e. it represents the number of times $a_1, \dots, a_n$ are used so far in solving the training puzzles of ARC-AGI1.

4. CONSTRUCTION OF THE TRAINING CEF (“THIS CORRESPONDS TO THIS”)

- (1) The main aim of these CEF is to identify on what interacting entities we have to focus our attention, find their abstract descriptions and order in the puzzle-solving algorithm (see “Define CEF using characterization”).